

JUNIK BAE

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EDUCATION

Seoul National University

Bachelor of Computer Science and Engineering, Minor in Mathematics
GPA: 4.12 / 4.30 (Cumulative), 4.13 / 4.30 (Major), 4.23 / 4.30 (Minor)
Leave of absence for military service: Feb 2021 - Nov 2022

2019 - Present
Seoul, South Korea

PUBLICATIONS

- [P1] [TLDR: Unsupervised Goal-Conditioned RL via Temporal Distance-Aware Representations](#)
Junik Bae, Kwanyoung Park, Youngwoon Lee
Conference on Robot Learning (CoRL), 2024
- [P2] [Exploiting Semantic Reconstruction to Mitigate Hallucinations in Vision-Language Models](#)
Minchan Kim*, Minyeong Kim*, **Junik Bae***, Suhwan Choi, Sungkyung Kim, Buru Chang
(*: equal contribution)
European Conference on Computer Vision (ECCV), 2024

AWARDS AND HONORS

- [C1] **1st Place**, [PokeAgent Challenge: RPG Speedrunning Track \(NeurIPS 2025\)](#) Oct 2025 - Nov 2025
Award: \$1,500 + \$700 (GCP credits), competition report co-authorship
Task: Developing an autonomous Pokemon Emerald speedrun agent for long-horizon decision-making
Method: Voyager-style executable skill generation + RL pipeline
- [C2] **1st Place**, [2022 Military AI Competition](#) Nov 2022 - Dec 2022
Awarded by Korean Minister of Science and Technology, ₩20,000,000 ($\approx$ \$15,000)
Preliminary Task: Change detection on buildings in aerial image data
Final Task: Image denoising for all-weather operations
- [C3] **2nd Place**, [2022 Korean AI Competition](#) Aug 2022 - Sep 2022
Awarded by Korean Minister of Science and Technology, ₩10,000,000 ($\approx$ \$7,500)
Task: Speech recognition on Korean dialect speech datasets
- [C4] **2nd Prize**, [Product Recognition Challenge on Self-service Stand](#) Sep 2021 - Oct 2021
Awarded by Chairman of Electrical and Computer Engineering Department at SNU
Task: High-precision-and-speed object detection on self-service stand images
- **SNU Semiconductor Track Scholarship** Mar 2024 - Present
Scholarship of ₩10,000,000 ($\approx$ \$7,500) to be conferred upon graduation

RESEARCH INTERESTS

My long-term goal is to develop exploration-driven, creative agents that can achieve superhuman performance in open-ended environments. In particular, I am interested in:

- **General unsupervised exploration algorithms** (e.g., [P1])
- **Knowledge-guided, sample-efficient exploration** (e.g., [C1])
- **LLM-based reasoning and agents** (e.g., [C1], [P2])

EXPERIENCE

Yonsei RLLAB Research Intern

Advisor: Prof. Youngwoon Lee

Mar 2024 - Present

- Researched and developed unsupervised goal-conditioned RL algorithm leveraging *temporal distance* for both exploration and goal-reaching [P1].

- Built a knowledge-guided RL pipeline using LLM-derived priors for long-horizon exploration, achieving first place in the NeurIPS 2025 PokeAgent Challenge (speedrun track) [C1].

SNU Deep Learning Research Club Member (13th Cohort)

Sep 2023 - Present

- Co-developed framework to mitigate hallucinations in vision-language models using semantic image reconstruction and reinforcement learning [P2].

SNU Vision and Learning Lab Research Intern

Jul 2023 - Sep 2023

Advisor: Prof. Gunhee Kim

- Developed face generation and video scene segmentation pipeline used in MBC entertainment show.
- Researched lifelong evaluation pipelines for retrieval-augmented generation models on dynamic text data streams.

Naver Cloud Speech Synthesis Team Research Intern

Jan 2023 - Feb 2023

- Designed and implemented text-to-speech (TTS) model tailored for proprietary Korean speech data.

Republic of Korea Air Force Software Development Specialist

Feb 2021 - Nov 2022

- Developed AI models to detect prohibited items in X-ray images.
- Built ML pipelines for gunshot detection and scoring during shooting training.
- Developed ML forecasting pipeline to predict munitions demand.

PROJECTS

- **TTS Model Implementation. (450+ stars)** (Github link) Implemented *NaturalSpeech: End-to-End Text to Speech Synthesis with Human-Level Quality* by Microsoft, former state-of-the-art model on LJ Speech Dataset. This is the first and only public implementation to the best of my knowledge.